Measure Theory and Integration

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1 Preliminaries

Definition 1.1 (Sequence)**.** Let X be a set. A *sequence* on X is a function $a : \mathbb{N} \to X$. The n-th term of the sequence a is denoted by a_n, i.e., $a_n \coloneqq a(n)$.

Definition 1.2 (Extended nonnegative real numbers)**.** We define

$$[0, \infty] \coloneqq \{x \in \mathbb{R} \mid x \geq 0\} \sqcup \{\infty\}.$$

We extend the order and the addition on the reals to $[0, \infty]$ by

$$a \leq \infty, \quad a + \infty = \infty + a = \infty$$

for any $a \in [0, \infty]$. For $a \in \mathbb{R}$, we define $a^+ \coloneqq \max(a, 0) \in [0, \infty]$.

Proposition 1.3 (The "ε-technique")**.** Let $a, b \in [0, \infty]$. To prove $a \leq b$, it suffices to show that $a \leq b + \varepsilon$ for any positive real $\varepsilon > 0$.

Proposition 1.4 (Countable decomposition of ε)**.** Let $\varepsilon > 0$. There is a sequence $\varepsilon' : \mathbb{N} \to \mathbb{R}$ such that $\varepsilon'_n > 0$ for all $n \in \mathbb{N}$ and

$$\sum_{n \in \mathbb{N}} \varepsilon'_n < \varepsilon.$$

2 Measure and integral on $\mathbb{R}$

2.1 Measure on $\mathbb{R}$

Definition 2.1 (Lebesgue measure)**.** Let $A \subseteq \mathbb{R}$. The *Lebesgue (outer) measure* of A is defined by

$$m(A) \coloneqq \inf_{a:\mathbb{N}\to\mathbb{R},\, b:\mathbb{N}\to\mathbb{R},\, A \subseteq \bigcup_n (a_n, b_n)} \sum_{n \in \mathbb{N}} (b_n - a_n)^+ \ \in \ [0, \infty].$$

Proposition 2.2 (Measure of closed intervals)**.** Let $a, b \in \mathbb{R}$. We have

$$m([a, b]) = (b - a)^+.$$

Proposition 2.3 (m is an outer measure)**.**

(1) $m(\emptyset) = 0$.

(2) For $A, B \subseteq \mathbb{R}$ with $A \subseteq B$, we have $m(A) \leq m(B)$.

(3) For $A_0, A_1, \cdots \subseteq \mathbb{R}$, we have

$$m\left(\bigcup_{n\in\mathbb{N}} A_n\right) \leq \sum_{n\in\mathbb{N}} m(A_n).$$

Remark 2.4. Let $A, B \subseteq \mathbb{R}$. It is known that, even if $A \cap B = \emptyset$, it does not necessarily imply that $m(A \cup B) = m(A) + m(B)$.

2.2 Measurable sets on $\mathbb{R}$

Definition 2.5 (Borel measurable set)**.** We define inductively the notion of a subset $A \subseteq \mathbb{R}$ being *Borel measurable* as follows:

(1) For $a \in \mathbb{R}$, the interval $(a, \infty]$ is Borel measurable.

(2) The empty set $\emptyset$ is Borel measurable.

(3) For a Borel measurable set A, its complement A^c is Borel measurable.

(4) For a sequence of Borel measurable sets $A_0, A_1, \ldots$, the union $\bigcup_{n\in\mathbb{N}} A_n$ is Borel measurable.

In what follows, measurable sets on $\mathbb{R}$ will mean Borel measurable sets.

Proposition 2.6 (Induction principle of Borel measurable sets)**.** Let $P(A)$ be a property on measurable sets $A \subseteq \mathbb{R}$. Suppose that the following conditions hold:

(1) For any $a \in \mathbb{R}$, we have $P((a, \infty])$.

(2) We have $P(\emptyset)$.

(3) For any measurable set $A \subseteq \mathbb{R}$, we have $P(A)$ implies $P(A^c)$.

(4) For any sequence of measurable sets $A_0, A_1, \cdots \subseteq \mathbb{R}$, we have $\forall n \in \mathbb{N}, P(A_n)$ implies $P\left(\bigcup_{n\in\mathbb{N}} A_n\right)$.

Then any measurable set $A \subseteq \mathbb{R}$ satisfies $P(A)$.

Definition 2.7. We say that a set $A \subseteq \mathbb{R}$ satisfies the Carathéodory's criterion if for any $B \subseteq \mathbb{R}$, we have

$$m(B) = m(B \cap A) + m(B \setminus A).$$

Proposition 2.8. Let $A \subseteq \mathbb{R}$. If A is measurable, then A satisfies the Carathéodory's criterion.

Proposition 2.9 (Countable additivity)**.** Let $A_0, A_1, \cdots \subseteq \mathbb{R}$ be a sequence of pairwise disjoint sets. Suppose that each A_n is measurable. We have

$$m\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \sum_{n\in\mathbb{N}} m(A_n).$$

Proposition 2.10 (Measure of arbitrary sets via measurable sets)**.** Let $A \subseteq \mathbb{R}$. We have

$$m(A) = \inf_{\substack{B\subseteq\mathbb{R}, \\ B \text{ is measurable}, A\subseteq B}} m(B).$$

Proposition 2.11. Let $A \subseteq \mathbb{R}$. There exists a measurable set $B \subseteq \mathbb{R}$ such that $A \subseteq B$ and $m(A) = m(B)$.

Proposition 2.12 (Continuity of measure from below)**.** Let $A_0, A_1, \cdots \subseteq \mathbb{R}$. Suppose that the following conditions hold:

(1) For any $n \in \mathbb{N}$, the set A_n is measurable.

(2) For any $i, j \in \mathbb{N}$ with $i \leq j$, we have $A_i \subseteq A_j$.

Then we have

$$m\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \sup_{n\in\mathbb{N}} m(A_n).$$

2.3 Integral of nonnegative functions

Definition 2.13 (Simple function)**.** A function $f : \mathbb{R} \to [0, \infty]$ is called *simple* if the following conditions hold:

(1) $f^{-1}\{x\}$ is measurable for any $x \in [0, \infty]$.

(2) The image of f is a finite set.

Definition 2.14 (Integral of simple functions)**.** Let $f : \mathbb{R} \to [0, \infty]$ be a simple function. We define the integral of f by

$$\text{Simp}\int_{x\in\mathbb{R}} f(x)\,dx \coloneqq \sum_{y\in\text{Im}(f)} y \cdot m(f^{-1}\{y\}) \quad \in [0, \infty].$$

Proposition 2.15 (Basic properties of the integral of simple functions)**.** Let $f, g : \mathbb{R} \to [0, \infty]$ be simple functions, let $c \in [0, \infty]$, and let $A \subseteq \mathbb{R}$ be measurable. Define a function $f|_A : \mathbb{R} \to [0, \infty]$ by

$$f|_A(x) \coloneqq \begin{cases} f(x) & \text{if } x \in A, \\ 0 & \text{if } x \notin A. \end{cases}$$

Then the following hold:

(1) The function $f|_A$ is simple.

(2) We have

$$\operatorname{Simp}\int_{x\in\mathbb{R}} f|_A(x)\,dx = \sum_{y\in\operatorname{Im}(f)} y\cdot m\big(A\cap f^{-1}\{y\}\big).$$

(3) We have

$$\operatorname{Simp}\int_{x\in\mathbb{R}} c\,dx = c\cdot m(\mathbb{R}).$$

(4) We have

$$\operatorname{Simp}\int_{x\in\mathbb{R}} c\,f(x)\,dx = c\cdot\operatorname{Simp}\int_{x\in\mathbb{R}} f(x)\,dx.$$

(5) We have

$$\operatorname{Simp}\int_{x\in\mathbb{R}} (f(x)+g(x))\,dx = \operatorname{Simp}\int_{x\in\mathbb{R}} f(x)\,dx + \operatorname{Simp}\int_{x\in\mathbb{R}} g(x)\,dx.$$

Definition 2.16 (Lower Lebesgue integral)**.** Let $f:\mathbb{R}\to[0,\infty]$ be a function. We define the *lower Lebesgue integral* of f by

$$\underline{\int}_{x\in\mathbb{R}} f(x)\,dx \coloneqq \sup_{\substack{g:\mathbb{R}\to[0,\infty],\\ g\text{ is simple},\, g\le f}} \operatorname{Simp}\int_{x\in\mathbb{R}} g(x)\,dx \quad \in[0,\infty].$$

Definition 2.17 (Measurable function)**.** Let $f:\mathbb{R}\to[0,\infty]$. We say that f is *measurable* if the set $f^{-1}(a,\infty]=\{x\mid a<f(x)\}$ is measurable for any $a\in[0,\infty]$.

For functions $f,g:\mathbb{R}\to[0,\infty]$, we write $f\le g$ if $f(x)\le g(x)$ for any $x\in\mathbb{R}$.

Theorem 2.18 (Monotone convergence theorem)**.** Let $f_0,f_1,\cdots:\mathbb{R}\to[0,\infty]$. Suppose that the following conditions hold:

(1) For any $n\in\mathbb{N}$, the function f_n is measurable.

(2) For any $i,j\in\mathbb{N}$ with $i\le j$, we have $f_i\le f_j$.

Then we have

$$\underline{\int}_{x\in\mathbb{R}} \sup_{n\in\mathbb{N}} f_n(x)\,dx = \sup_{n\in\mathbb{N}} \underline{\int}_{x\in\mathbb{R}} f_n(x)\,dx.$$

Proposition 2.19. Take a bijection $q:\mathbb{N}\to\mathbb{Q}_{\ge 0}$. Let $f:\mathbb{R}\to[0,\infty]$ be measurable. For $n\in\mathbb{N}$, define $\operatorname{approx}_n f:\mathbb{R}\to[0,\infty]$ by

$$(\operatorname{approx}_n f)(x) \coloneqq \max_{\substack{0\le i<n\\ q_i\le f(x)}} q_i.$$

Then we have the following properties:

(1) For any n, the function $\operatorname{approx}_n f$ is simple.

(2) For any i, j with $i \leq j$, we have $\operatorname{approx}_i f \leq \operatorname{approx}_j f$.

(3) For any $x \in \mathbb{R}$, we have

$$\sup_{n \in \mathbb{N}} (\operatorname{approx}_n f)(x) = f(x).$$

Corollary 2.20. Let $f : \mathbb{R} \to [0, \infty]$ be measurable. For a bijection $q : \mathbb{N} \to \mathbb{Q}_{\geq 0}$, define $\operatorname{approx}_n f$ as above. Then

$$\underline{\int}_{x \in \mathbb{R}} f(x)\, dx = \sup_{n \in \mathbb{N}} \operatorname{Simp} \int_{x \in \mathbb{R}} (\operatorname{approx}_n f)(x)\, dx.$$

Proposition 2.21. Let $f, g : \mathbb{R} \to [0, \infty]$ be measurable functions. We have

$$\underline{\int}_{x \in \mathbb{R}} f(x) + g(x)\, dx = \underline{\int}_{x \in \mathbb{R}} f(x)\, dx + \underline{\int}_{x \in \mathbb{R}} g(x)\, dx.$$

If in addition $g \leq f$, we have

$$\underline{\int}_{x \in \mathbb{R}} f(x) - g(x)\, dx = \underline{\int}_{x \in \mathbb{R}} f(x)\, dx - \underline{\int}_{x \in \mathbb{R}} g(x)\, dx.$$

Definition 2.22. For a sequence $a : \mathbb{N} \to [0, \infty]$, the *limit infimum* and the *limit supremum* of a are defined by

$$\liminf_{n \to \infty} a_n \coloneqq \sup_{n \in \mathbb{N}} \inf_{i \geq n} a_i, \quad \limsup_{n \to \infty} a_n \coloneqq \inf_{n \in \mathbb{N}} \sup_{i \geq n} a_i.$$

Lemma 2.23. Let $a : \mathbb{N} \to [0, \infty]$. We have $\liminf_{n \to \infty} a_n \leq \limsup_{n \to \infty} a_n$.

Lemma 2.24. Let $u : \mathbb{N} \to [0, \infty]$ and $a \in [0, \infty]$. If

$$a \leq \liminf_{n \to \infty} u_n \quad \text{and} \quad \limsup_{n \to \infty} u_n \leq a.$$

then $\lim_{n \to \infty} u_n = a$.

Theorem 2.25 (Fatou's lemma)**.** Let $f_0, f_1, \cdots : \mathbb{R} \to [0, \infty]$ be a sequence of measurable functions. We have

$$\underline{\int}_{x \in \mathbb{R}} \liminf_{n \to \infty} f_n(x)\, dx \leq \liminf_{n \to \infty} \underline{\int}_{x \in \mathbb{R}} f_n(x)\, dx.$$

Theorem 2.26 (Dominated convergence theorem)**.** Let $f_0, f_1, \cdots : \mathbb{R} \to [0, \infty]$ be a sequence of measurable functions, and let $F : \mathbb{R} \to [0, \infty]$ be a function. Let $g : \mathbb{R} \to [0, \infty]$ be another function, and suppose that the following conditions hold:

(1) For any n and for any $x \in \mathbb{R}$, we have $f_n(x) \leq g(x)$.

(2) $\underline{\int}_{x\in\mathbb{R}} g(x)\,dx < \infty$.

(3) For any $x \in \mathbb{R}$, the sequence $f_0(x), f_1(x), \dots$ converges to $F(x)$.

Then we have

$$\lim_{n\to\infty} \underline{\int}_{x\in\mathbb{R}} f_n(x)\,dx = \underline{\int}_{x\in\mathbb{R}} F(x)\,dx.$$

2.4 Integral of $\mathbb{R}$-valued functions

Definition 2.27. Let $f : \mathbb{R} \to \mathbb{R}$. We say that f is *measurable* if the set $f^{-1}(a, \infty) = \{x \mid a < f(x)\}$ is measurable for any $a \in \mathbb{R}$.

Definition 2.28. Let $f : \mathbb{R} \to \mathbb{R}$. We say that f is *integrable* if the following conditions hold:

(1) f is measurable.

(2) We have

$$\underline{\int}_{x\in\mathbb{R}} |f(x)|\,dx < \infty.$$

Definition 2.29. Let $f : \mathbb{R} \to \mathbb{R}$ be an integrable function. We define the Lebesgue integral of f by

$$\int_{x\in\mathbb{R}} f(x)\,dx \coloneqq \underline{\int}_{x\in\mathbb{R}} f_+(x)\,dx - \underline{\int}_{x\in\mathbb{R}} f_-(x)\,dx \quad \in \mathbb{R},$$

where

$$f_+(x) \coloneqq \max(f(x), 0), \quad f_-(x) \coloneqq \max(-f(x), 0).$$

Theorem 2.30 (Dominated convergence theorem)**.** Let $f_0, f_1, \dots : \mathbb{R} \to \mathbb{R}$ be a sequence of measurable functions, and let $F : \mathbb{R} \to \mathbb{R}$ be a function. Let $g : \mathbb{R} \to \mathbb{R}$ be another function, and suppose that the following conditions hold:

(1) For any n and for any $x \in \mathbb{R}$, we have $|f_n(x)| \le g(x)$.

(2) g is integrable.

(3) For any $x \in \mathbb{R}$, the sequence $f_0(x), f_1(x), \dots$ converges to $F(x)$.

Then we have

$$\lim_{n\to\infty} \int_{x\in\mathbb{R}} f_n(x)\,dx = \int_{x\in\mathbb{R}} F(x)\,dx.$$

3 Measure theory

3.1 Integral with respect to a measure

For a set X, we denote by $\mathcal{P}(X)$ the power set of X, i.e., the set of all subsets of X.

Definition 3.1 (Measurable space)**.** Let X be a set. A *sigma-algebra* on X is a set $\mathcal{M} \subseteq \mathcal{P}(X)$ satisfying the following conditions:

(1) $\emptyset \in \mathcal{M}$.

(2) For any $A \in \mathcal{M}$, its complement A^c is in $\mathcal{M}$.

(3) For any sequence $A_0, A_1, \cdots \in \mathcal{M}$, the union $\bigcup_{n\in\mathbb{N}} A_n$ is in $\mathcal{M}$.

A set X equipped with a sigma-algebra $\mathcal{M}$ is called a *measurable space*. In this case, a set $A \subseteq X$ is called *measurable* if $A \in \mathcal{M}$.

When X is a measurable space, the set of measurable subsets of X is often denoted by $\mathcal{M}(X)$.

Definition 3.2 (Measure)**.** Let X be a measurable space. A *measure* is a function $\mu : \mathcal{M}(X) \to [0, \infty]$ satisfying the following properties:

(1) $\mu(\emptyset) = 0$.

(2) For any pairwise disjoint sequence of measurable sets $A_0, A_1, \cdots \subseteq X$, we have
$$\mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \sum_{n\in\mathbb{N}} \mu(A_n).$$

Definition 3.3 (Measurable function)**.** Let X and Y be measurable spaces. A function $f : X \to Y$ is said to be *measurable* if for any measurable set $B \subseteq Y$, the set $f^{-1}(B) \subseteq X$ is measurable.

Definition 3.4 (Simple function)**.** Let X be a measurable space, and let Y be a set. A function $f : X \to Y$ is called *simple* if the following conditions hold:

(1) $f^{-1}\{y\}$ is measurable for any $y \in Y$

(2) The image of f is a finite set.

Definition 3.5 (Integral of simple functions)**.** Let X be a measurable space, and let μ be a measure on X. Let $f : X \to [0, \infty]$ be a simple function. We define the integral of f with respect to μ by
$$\mathrm{Simp}\int_{x\in X} f(x)\, d\mu := \sum_{y\in \mathrm{Im}(f)} y \cdot \mu(f^{-1}\{y\}).$$

Definition 3.6 (Lower Lebesgue integral)**.** Let X be a measurable space, and let μ be a measure on X. Let $f : X \to [0, \infty]$ be a function. We define the *lower Lebesgue integral* of f with respect to μ by

$$\underline{\int}_{x \in X} f(x)\, d\mu \coloneqq \sup_{\substack{g : X \to [0,\infty], \\ g \text{ is simple}, g \le f}} \operatorname{Simp} \int_{x \in X} g(x)\, d\mu.$$

Theorem 3.7 (Monotone convergence theorem)**.** Let X be a measurable space, and let μ be a measure on X. Let $f_0, f_1, \cdots : X \to [0, \infty]$ be a sequence of measurable functions. Suppose that $f_i \le f_j$ if $i \le j$. Then we have

$$\underline{\int}_{x \in X} \sup_{n \in \mathbb{N}} f_n(x)\, d\mu = \sup_{n \in \mathbb{N}} \underline{\int}_{x \in X} f_n(x)\, d\mu.$$

Theorem 3.8 (Fatou's lemma)**.** Let X be a measurable space, and let μ be a measure on X. Let $f_0, f_1, \cdots : X \to [0, \infty]$ be a sequence of measurable functions. We have

$$\underline{\int}_{x \in X} \liminf_{n \to \infty} f_n(x)\, d\mu \le \liminf_{n \to \infty} \underline{\int}_{x \in X} f_n(x)\, d\mu.$$

Theorem 3.9 (Dominated convergence theorem)**.** Let X be a measurable space, and let μ be a measure on X. Let $f_0, f_1, \cdots : X \to [0, \infty]$ be a sequence of measurable functions, and let $F : X \to [0, \infty]$ be a function. Let $g : X \to [0, \infty]$ be another function, and suppose that the following conditions hold:

(1) For any n and for any $x \in X$, we have $f_n(x) \le g(x)$.

(2) $\underline{\int}_{x \in X} g(x)\, d\mu < \infty$.

(3) For any $x \in X$, the sequence $f_0(x), f_1(x), \ldots$ converges to $F(x)$.

Then we have

$$\lim_{n \to \infty} \underline{\int}_{x \in X} f_n(x)\, d\mu = \underline{\int}_{x \in X} F(x)\, d\mu.$$

Definition 3.10. Let $f : X \to \mathbb{R}$. We say that f is *integrable* with respect to μ if the following conditions hold:

(1) f is measurable.

(2) We have

$$\underline{\int}_{x \in X} |f(x)|\, d\mu < \infty.$$

Definition 3.11. Let $f : X \to \mathbb{R}$ be an integrable function. We define the Lebesgue integral of f with respect to μ by

$$\int_{x \in X} f(x)\, d\mu \coloneqq \underline{\int}_{x \in X} f_+(x)\, d\mu - \underline{\int}_{x \in X} f_-(x)\, d\mu \quad \in \mathbb{R},$$

where

$$f_+(x) \coloneqq \max(f(x), 0), \quad f_-(x) \coloneqq \max(-f(x), 0).$$

3.2 Fundamental theorem of calculus

Definition 3.12 (Restriction of a measure)**.** Let X be a measurable space, and let μ be a measure on X. Let $A \subseteq X$ be a measurable set. We define the *restriction of μ to A*, denoted by $\mu|_A$, as the measure on X defined by

$$\mu|_A(B) \coloneqq \mu(A \cap B)$$

for any measurable set $B \subseteq X$.

Definition 3.13. Let $f : X \to \mathbb{R}$, and let $a, b \in \mathbb{R}$. The *interval integral* of f from a to b is defined by

$$\int_a^b f(x)\,dx \coloneqq \begin{cases} \displaystyle\int_{x\in\mathbb{R}} f(x)\,d(\mu|_{(a,b]}) & \text{if } a \le b, \\ \displaystyle -\int_{x\in\mathbb{R}} f(x)\,d(\mu|_{(b,a]}) & \text{if } b \le a, \end{cases}$$

when both integrals on the right-hand side are integrable.

Theorem 3.14. Let $f : \mathbb{R} \to \mathbb{R}$ and $a \in \mathbb{R}$. Suppose that f is measurable and $\lim_{x\to a} f(x) = f(a)$. Then we have

$$\frac{d}{dt}\left(\int_a^t f(x)\,dx\right)\bigg|_{t=a} = c.$$

Corollary 3.15. Let $f, F : \mathbb{R} \to \mathbb{R}$ and $a, b \in \mathbb{R}$. Suppose that f is continuous on $[a, b]$, and that $f(x) = F'(x)$ for any $x \in [a, b]$. Then we have

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

3.3 Outer measure

Definition 3.16 (Outer measure associated with a measure)**.** Let X be a measurable space, and let μ be a measure on X. We define a function $\mu^* : \mathcal{P}(X) \to [0, \infty]$ by

$$\mu^*(A) \coloneqq \inf_{\substack{B\subseteq X, \\ B \text{ is measurable}, A\subseteq B}} \mu(B)$$

for any $A \subseteq X$.

Lemma 3.17. Let X be a measurable space, and let μ be a measure on X.

(1) For any $A \subseteq X$, we have $\mu(A) \le \mu^*(A)$.

(2) For any measurable set $A \subseteq X$, we have $\mu(A) = \mu^*(A)$.

(3) For any $A \subseteq X$, there exists a set $B \subseteq X$ such that

- B is measurable,

- $A \subseteq B$,
- $\mu^*(A) = \mu(B)$.

Lemma 3.18. Let X be a measurable space, and let μ be a measure on X.

(1) $\mu^*(\emptyset) = 0$.

(2) For any $A, B \subseteq X$ with $A \subseteq B$, we have $\mu^*(A) \leq \mu^*(B)$.

(3) For any sequence of sets $A_0, A_1, \cdots \subseteq X$, we have

$$\mu^*\left(\bigcup_{n\in\mathbb{N}} A_n\right) \leq \sum_{n\in\mathbb{N}} \mu^*(A_n).$$

Definition 3.19 (Outer measure)**.** Let X be a set. An *outer measure* on X is a function $\mu : \mathcal{P}(X) \to [0, \infty]$ satisfying the following properties:

(1) $\mu(\emptyset) = 0$.

(2) For $A, B \subseteq X$ with $A \subseteq B$, we have $\mu(A) \leq \mu(B)$.

(3) For any sequence of sets $A_0, A_1, \cdots \subseteq X$, we have

$$\mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) \leq \sum_{n\in\mathbb{N}} \mu(A_n).$$

Definition 3.20 (Carathéodory' criterion)**.** Let X be a set, and let μ be an outer measure on X. We say that a set $A \subseteq X$ is *carathéodory* with respect to μ if for any set $B \subseteq X$, we have

$$\mu(B) = \mu(B \cap A) + \mu(B \setminus A).$$

Lemma 3.21. Let X be a measurable space, and let μ be a measure on X. Then any measurable set $A \subseteq X$ is carathéodory with respect to μ^*.

Proposition 3.22 (Measure from outer measure)**.** Let X be a measurable space, and let $\bar{\mu}$ be an outer measure on X. Suppose that any measurable set $A \subseteq X$ is carathéodory with respect to $\bar{\mu}$. Define a function $\mu : \mathcal{M}(X) \to [0, \infty]$ by

$$\mu(A) \coloneqq \bar{\mu}(A)$$

for any measurable set $A \subseteq X$. Then μ is a measure on X. We call μ the *measure associated with* $\bar{\mu}$.

3.4 Measurable space generated by a set of sets

Definition 3.23 (Measurable space generated by a set of sets)**.** Let X be a set, and let $\mathcal{F} \subseteq \mathcal{P}(X)$. We inductively define the sigma-algebra $\mathsf{gen}(\mathcal{F})$ on X, called the *sigma-algebra generated by* $\mathcal{F}$, as follows:

(1) For any $A \in \mathcal{F}$, we have $A \in \mathsf{gen}(\mathcal{F})$.

(2) $\emptyset \in \mathsf{gen}(\mathcal{F})$.

(3) For any $A \in \mathsf{gen}(\mathcal{F})$, we have $A^c \in \mathsf{gen}(\mathcal{F})$.

(4) For any sequence $A_0, A_1, \cdots \in \mathsf{gen}(\mathcal{F})$, we have $\bigcup_{n\in\mathbb{N}} A_n \in \mathsf{gen}(\mathcal{F})$.

Definition 3.24 (Borel measurable set)**.** Let X be a topological space (or a metric space, if you are not familiar with topological spaces). Then we regard X as a measurable space, equipping it with the sigma-algebra $\mathsf{gen}(\{U \subseteq X \mid U \text{ is open}\})$. In this situation, a measurable set $A \subseteq X$ is called a *Borel measurable set*.

Proposition 3.25 (Induction principle for measurable sets)**.** Let X be a measurable space, and let $\mathcal{F} \subseteq \mathcal{P}(X)$. Suppose that $\mathcal{M}(X) = \mathsf{gen}(\mathcal{F})$. Let $P(A)$ be a property on measurable sets $A \subseteq X$. Suppose that the following conditions hold:

(1) For any $A \in \mathcal{F}$, we have $P(A)$.

(2) We have $P(\emptyset)$.

(3) For any measurable set $A \subseteq X$, we have $P(A)$ implies $P(A^c)$.

(4) For any sequence of measurable sets $A_0, A_1, \cdots \subseteq X$, we have $\forall n \in \mathbb{N}, P(A_n)$ implies $P\left(\bigcup_{n\in\mathbb{N}} A_n\right)$.

Then any measurable set $A \subseteq X$ satisfies $P(A)$.

Definition 3.26. Let X be a set, and let $\mathcal{F} \subseteq \mathcal{P}(X)$. We inductively define the set of sets $\mathsf{d\text{-}gen}(\mathcal{F}) \subseteq \mathcal{P}(X)$, called the *Dynkin system generated by* $\mathcal{F}$, as follows:

(1) For any $A \in \mathcal{F}$, we have $A \in \mathsf{d\text{-}gen}(\mathcal{F})$.

(2) $\emptyset \in \mathsf{d\text{-}gen}(\mathcal{F})$.

(3) For any $A \in \mathsf{d\text{-}gen}(\mathcal{F})$, we have $A^c \in \mathsf{d\text{-}gen}(\mathcal{F})$.

(4) For any sequence $A_0, A_1, \cdots \in \mathsf{d\text{-}gen}(\mathcal{F})$ such that $A_i \cap A_j = \emptyset$ if $i \neq j$, we have $\bigcup_{n\in\mathbb{N}} A_n \in \mathsf{d\text{-}gen}(\mathcal{F})$.

Proposition 3.27 (Dynkin's π-λ theorem)**.** Let X be a measurable space, and let $\mathcal{F} \subseteq \mathcal{P}(X)$. Suppose that $\mathcal{M}(X) = \mathsf{gen}(\mathcal{F})$. Suppose that for any $A, B \in \mathcal{F}$, we have $A \cap B \in \mathcal{F}$. Then for any $A \subseteq X$, A is measurable if and only if $A \in \mathsf{d\text{-}gen}(\mathcal{F})$.

Proposition 3.28 (Dinkin's induction principle for measurable sets)**.** Let X be a measurable space, and let $\mathcal{F} \subseteq \mathcal{P}(X)$. Suppose that $\mathcal{M}(X) = \mathsf{gen}(\mathcal{F})$. Suppose that for any $A, B \in \mathcal{F}$, we have $A \cap B \in \mathcal{F}$. Let $P(A)$ be a property on measurable sets $A \subseteq X$. Suppose that the following conditions hold:

(1) For any $A \in \mathcal{F}$, we have $P(A)$.

(2) We have $P(\emptyset)$.

(3) For any measurable set $A \subseteq X$, we have $P(A)$ implies $P(A^c)$.

(4) For any sequence of measurable sets $A_0, A_1, \cdots \subseteq X$ such that $A_i \cap A_j = \emptyset$ if $i \neq j$, we have $\forall n \in \mathbb{N}, P(A_n)$ implies $P\left(\bigcup_{n\in\mathbb{N}} A_n\right)$.

Then any measurable set $A \subseteq X$ satisfies $P(A)$.

3.5 Product measure

Definition 3.29 (Set of measures as a measurable space)**.** Let X be a measurable space. We denote by $\mathrm{Measure}(X)$ the set of all measures on X. We regard $\mathrm{Measure}(X)$ as a measurable space by equipping it with the sigma-algebra generated by the set

$$\{(\mu \mapsto \mu(A))^{-1}(B) \mid A \subseteq X \text{ is measurable and } B \subseteq [0, \infty] \text{ is measurable}\}.$$

Exercise 3.30. Let $A \subseteq X$ be a measurable set. Show that the function from $\mathrm{Measure}(X)$ to $[0, \infty]$ defined by $\mu \mapsto \mu(A)$ is measurable.

Proposition 3.31 (Kernel integration)**.** Let X and Y be measurable spaces. Let $\nu : X \to \mathrm{Measure}(Y)$. Suppose that ν is a measurable function. We have a function $\mathcal{K}_\nu : \mathrm{Measure}(X) \to \mathrm{Measure}(Y)$ defined by

$$\mathcal{K}_\nu(\mu)(A) = \int_{x\in X} \nu(x)(A)\, d\mu$$

for any measure μ on X and for any measurable set $A \subseteq Y$.

Definition 3.32 (Product of measurable spaces)**.** Let X and Y be measurable spaces. We regard the set $X \times Y$ as a measurable space by equipping it with the sigma-algebra generated by the set

$$\{A \times B \mid A \subseteq X \text{ and } B \subseteq Y \text{ are measurable}\}.$$

Definition 3.33. Let X be a measurable space. A measure μ on X is said to be *finite* if $\mu(X) < \infty$.

Proposition 3.34. Let X be a measurable space. Let $\mu : \mathbb{N} \to \mathrm{Measure}(X)$ be a sequence of measures on X. Then the function $\sum_{n\in\mathbb{N}} \mu_n$ defined by

$$\left(\sum_{n\in\mathbb{N}} \mu_n\right)(A) \coloneqq \sum_{n\in\mathbb{N}} \mu_n(A)$$

for any measurable set $A \subseteq X$ is a measure on X.

Definition 3.35 (S-finite)**.** Let X be a measurable space, and let μ be a measure on X. We say that μ is *S-finite* if there exists a sequence $\nu : \mathbb{N} \to \mathrm{Measure}(X)$ of measure on X such that ν_n is finite for any $n \in \mathbb{N}$ and $\mu = \sum_{n\in\mathbb{N}} \nu_n$.

Example 3.36. The Lebesgue measure on $\mathbb{R}$ is S-finite. This is because we have

$$m = \sum_{n\in\mathbb{Z}} m|_{(n,n+1]}.$$

Proposition 3.37. Let X and Y be measurable spaces. Let μ be a measure on X, and let ν be a measure on Y. Suppose that ν is S-finite. Then for any measurable sets $A \subseteq X \times Y$, the function $f : X \to [0,\infty]$ defined by $f(x) := \nu(\{y \in Y \mid (x,y) \in A\})$ is measurable. Suppose in addition that μ is also S-finite. Then for any measurable sets $A \subseteq X \times Y$, the function $g : Y \to [0,\infty]$ defined by $g(y) := \mu(\{x \in X \mid (x,y) \in A\})$ is measurable.

Definition 3.38 (Product measure)**.** Let X and Y be measurable spaces. Let μ be a measure on X, and let ν be a measure on Y. Suppose that ν is S-finite. Let $\nu' : X \to \mathrm{Measure}(X \times Y)$ be a function defined by

$$\nu'(x)(A) := \nu(\{y \in Y \mid (x,y) \in A\}).$$

We define the *product measure* $\mu \times \nu$ on $X \times Y$ by

$$\mu \times \nu := \mu\mathcal{K}_{\nu'}.$$

Exercise 3.39. Suppose that μ and ν are S-finite. Show that for any measurable set $A \subseteq X \times Y$, we have

$$\begin{aligned}(\mu \times \nu)(A) &= \underline{\int}_{x\in X} \nu(\{y \in Y \mid (x,y) \in A\})\, d\mu \\ &= \underline{\int}_{y\in Y} \mu(\{x \in X \mid (x,y) \in A\})\, d\nu.\end{aligned}$$

Exercise 3.40. Suppose that μ and ν are S-finite. Show that for any measurable sets $A \subseteq X$ and $B \subseteq Y$, we have

$$(\mu \times \nu)(A \times B) = \mu(A) \cdot \nu(B).$$

Theorem 3.41 (Tonelli's theorem)**.** Let X and Y be measurable spaces. Let μ be a measure on X, and let ν be a measure on Y. Suppose that μ and ν are S-finite. Let $f : X \times Y \to [0,\infty]$ be a measurable function. Then we have

$$\begin{aligned}\underline{\int}_{(x,y)\in X\times Y} f(x,y)\, d(\mu \times \nu) &= \underline{\int}_{x\in X} \left(\underline{\int}_{y\in Y} f(x,y)\, d\nu \right) d\mu \\ &= \underline{\int}_{y\in Y} \left(\underline{\int}_{x\in X} f(x,y)\, d\mu \right) d\nu.\end{aligned}$$